Sreyas Institute of Engineering and Technology

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DEEP LEARNING LAB PROJECT

on

Multi-Layer Perceptron for Bank Churn Prediction

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Abstract

- Bank customer churn refers to customers leaving a bank or closing their accounts. Predicting churn is important because acquiring new customers is more expensive than retaining existing ones.
- This project uses an Artificial Neural Network (ANN) with a Multi-Layer Perceptron (MLP) architecture to predict whether a customer will leave the bank. The model analyzes customer attributes such as age, credit score, account balance, and transaction behavior.
- By learning patterns in historical customer data, the neural network predicts the probability of churn. This helps banks take preventive actions such as offering promotions or improving services to retain customers

Introduction

- Customer churn prediction is a major problem in the banking industry. Banks lose significant revenue when customers close their accounts or move to competitors.
- Traditional statistical methods often fail to capture complex relationships between customer attributes and churn behavior. Machine learning and deep learning techniques provide more accurate predictions.
- Artificial Neural Networks are particularly useful because they can learn **non-linear patterns** in data. A **Multi-Layer Perceptron (MLP)** is a type of feed-forward neural network that contains multiple hidden layers to learn complex patterns.
- In this project, an ANN-based model is used to predict whether a bank customer will churn or stay with the bank.

Problem Statement

- Banks need an intelligent system to identify customers who are likely to leave the bank.
- Traditional analysis methods are not effective for large datasets and complex customer behaviors.
- The problem addressed in this project is to develop a **deep learning model using ANN (MLP)** that can accurately predict customer churn based on historical customer information.
- This predictive system helps banks reduce customer loss and improve customer retention strategies.

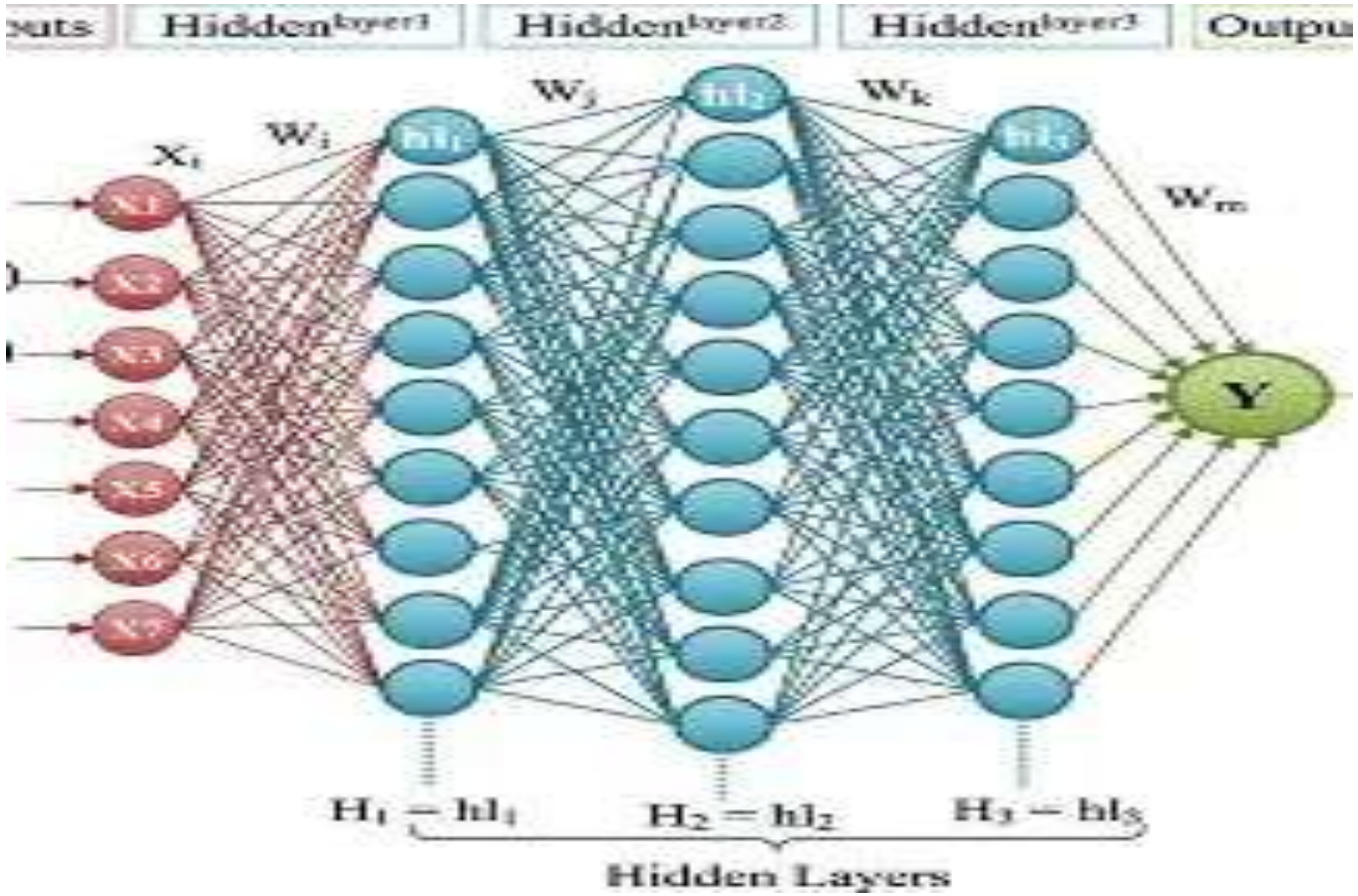
Dataset Description

- The dataset used in this project is the **Bank Customer Churn Dataset**.
- Typical features in the dataset include:
- • Customer ID
  • Credit Score
  • Geography
  • Gender
  • Age
  • Tenure (years with bank)
  • Account Balance
  • Number of Products
  • Has Credit Card
  • Active Member Status
  • Estimated Salary
  • Churn (Target Variable)
- The dataset contains thousands of customer records used to train and test the neural network model

Methodology/ Model Architecture

- The proposed system follows the following steps:
- Data Collection – Load customer dataset
- Data Preprocessing – Handle missing values and encode categorical variables
- Feature Scaling – Normalize numerical features
- Train-Test Split – Divide data into training and testing sets
- Model Building – Create ANN using Multi-Layer Perceptron
- Model Training – Train neural network with training data
- Prediction – Predict customer churn
- Evaluation – Measure model accuracy

Model Architecture



Code & Output

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
data = pd.read_csv("churn.csv")
X = data.drop("Exited", axis=1)
y = data["Exited"]
scaler = StandardScaler()
X = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2)
model = Sequential()
model.add(Dense(16, activation='relu', input_dim=X.shape[1]))
model.add(Dense(8, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
model.fit(X_train, y_train, epochs=20, batch_size=32)
```

Output:

```
Epoch 1/20
250/250
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s
3ms/step - loss: 0.62 - accuracy: 0.66
Epoch 2/20
250/250
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s
2ms/step - loss: 0.51 - accuracy: 0.74
...
Epoch 20/20
250/250
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s
2ms/step - loss: 0.42 - accuracy: 0.81
Model Loss: 0.41
Model Accuracy: 0.82
Sample Predictions:
[[False]
 [ True]
 [False]
 [False]
 [ True]]
```

Training

- During training, the neural network adjusts weights using **backpropagation** and **gradient descent**.
- Key training parameters:
- • Optimizer: Adam
  • Loss Function: Binary Crossentropy
  • Epochs: 20
  • Batch Size: 32
- The training process allows the neural network to learn patterns from customer data and improve prediction accuracy.

Results and Discussion

After training, the model predicts whether a customer will churn.

Evaluation metrics:

- Accuracy
- Precision
- Recall
- Confusion Matrix

Results show that the ANN model can effectively identify customers at risk of leaving the bank.

Higher prediction accuracy indicates that neural networks are effective for churn prediction problems.

Applications

- Customer churn prediction systems can be used in many industries:
- • Banking – Identify customers likely to leave
  • Telecommunications – Reduce subscriber loss
  • E-commerce – Retain online customers
  • Subscription services – Predict cancellation behavior
- Banks can use churn prediction models to design targeted retention strategies

Conclusion & Future Scope

- This project demonstrates the use of **Artificial Neural Networks with Multi-Layer Perceptron architecture** for predicting bank customer churn.
- The model learns patterns from historical data and predicts churn probability with good accuracy.
- Advantages:
- • Handles complex relationships in data
  • Improves customer retention strategies
  • Supports data-driven decision making
- Future improvements may include:
- • Using Deep Neural Networks
  • Applying feature selection techniques
  • Using advanced models like LSTM or XGBoost
  • Deploying the model as a web application

References

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